SET – A

Reg No.	1	4	5	1	1				

SATHYABAMA
INSTITUTE OF SCIENCE AND TECHNOLOGY
(DEEMED TO BE UNIVERSITY)

CONTINUOUS ASSESSMENT EXAM-1 (SET-A)

Programme: BE and B.Tech (except Bio and Specialization) Max. Marks: 50
Course Code: SMTB1201 Time: 2 Hours
Course Name: Advanced Calculus and Statistics Sem: II
Batch : 2025-2029 Date: 19.02.2026

Part-A Answer ALL the questions (7 × 2=14)

Q.No	Questions	CO
1.	Find the complementary function of $(D^2 + 2D + 1)y = 0$	1
2.	Find the PI of $(D^2 - 2D + 1)\, y = e^x$	1
3.	Convert $(x^2D^2 + xD + 1)y = \log x(\sin(\log x))$ in to differential equation with constant coefficients.	1
4.	Sove $(D^2 - 1)\, y = x$.	1
5.	Write the formula to find out the angle between two surfaces φ_1 and φ_2.	2
6.	Examine $\vec{F} = (z^2 + 2x + 3y)\,\vec{i} + (3x + 2y + z)\,\vec{j} + (y+2zx)\,\vec{k}$ is irrotational or not.	2
7.	Recall Stoke's theorem.	2

Part-B Answer ALL the questions (3 × 12 = 36)

8.	Solve $(D^2 + 4D + 4)\, y = e^{-3x} \sin 2x$	1
	(OR)	
9.	Solve $(x^2D^2 + 4xD + 2)\, y = \cos(\log x)$	1
10.	Solve $(D^2 + 4)\, y = 4 \tan 2x$, by the method of variation of parameters.	1
	(OR)	
11.	Prove that $\vec{F} = (y^2 \cos x + z^3)\,\vec{i} + (2y \sin x - 4)\,\vec{j} + 3xz^2\,\vec{k}$ is irrotational and find its scalar potential.	2

12.	Verify Stoke's theorem for $\vec{F} = x^2\vec{i} + xy\vec{j}$, integrated around the square in the z = 0 plane whose sides are along the lines x = 0, x = a, and y = 0, y = a	2
	(OR)	
13.	Verify Green's theorem for $\int_C [(x^2 + y)dx - xy^2 dy]$, where C is taken around the boundaries of the square whose vertices are (0,0), (0,1), (1,0), (1,1).	2